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W3 Lesson 1



DeepLearning.AI

Vectors and Linear Transformations

Machine Learning motivation

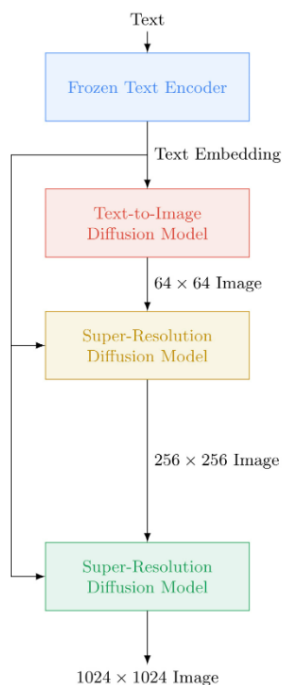
Neural Networks - AI generated images



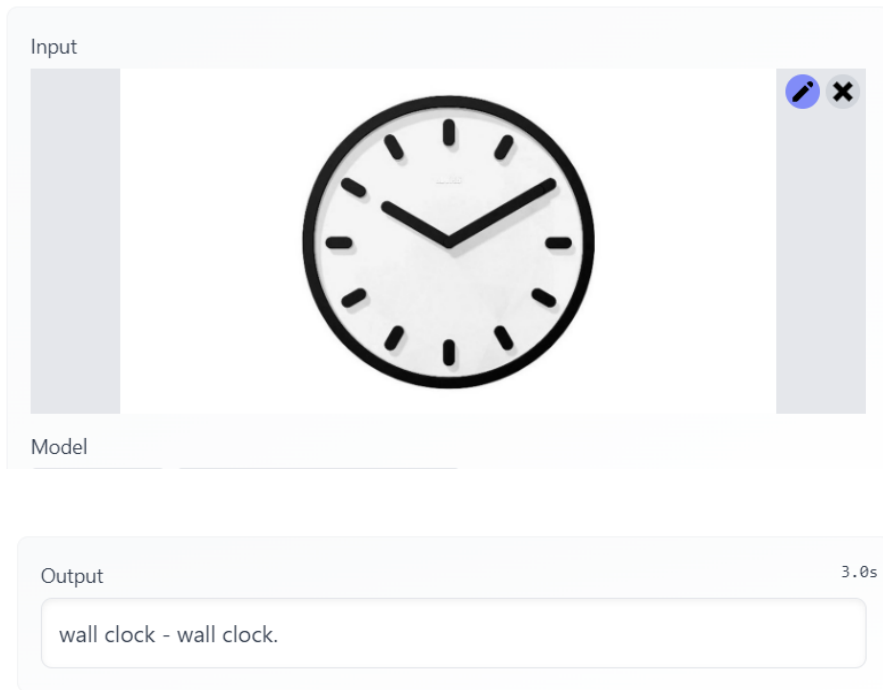
AI-generated human faces.

- Generative learning: Generating realistic looking images.

Text-to-image and image-to-text generation



"A Golden Retriever dog wearing a blue checkered beret and red dotted turtleneck."



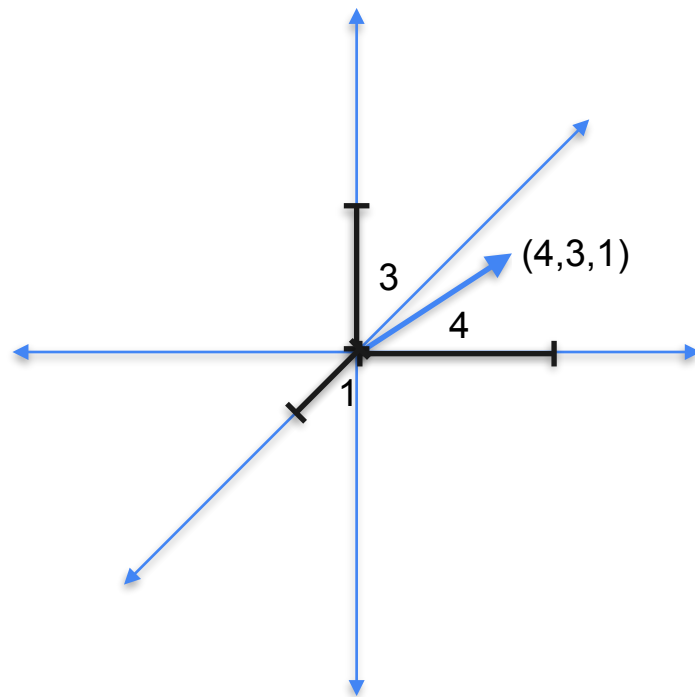
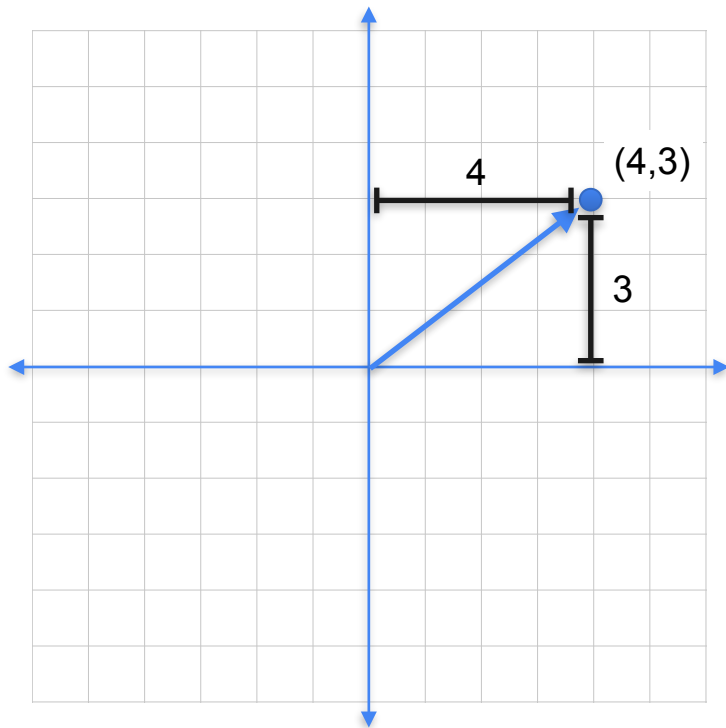


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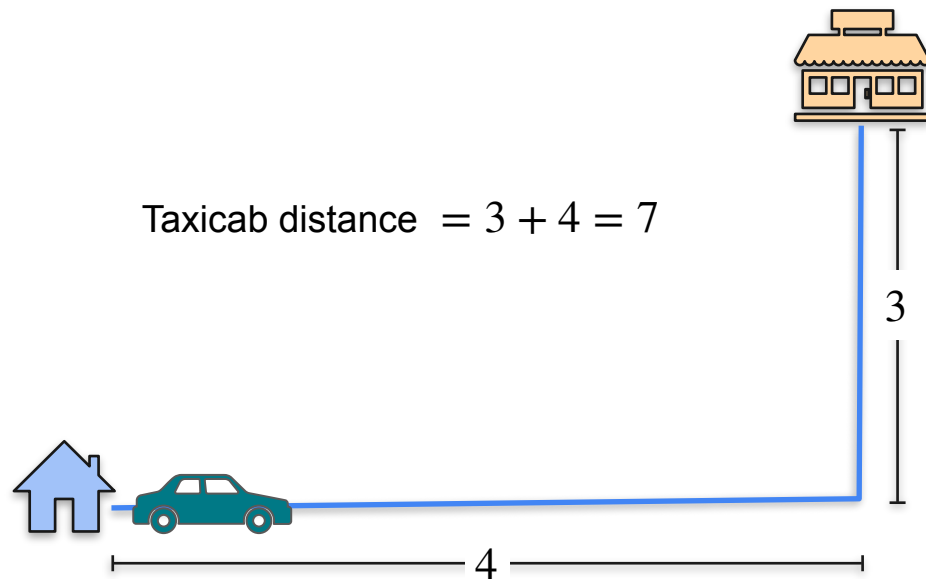
Vectors and Linear Transformations

Vectors and their properties

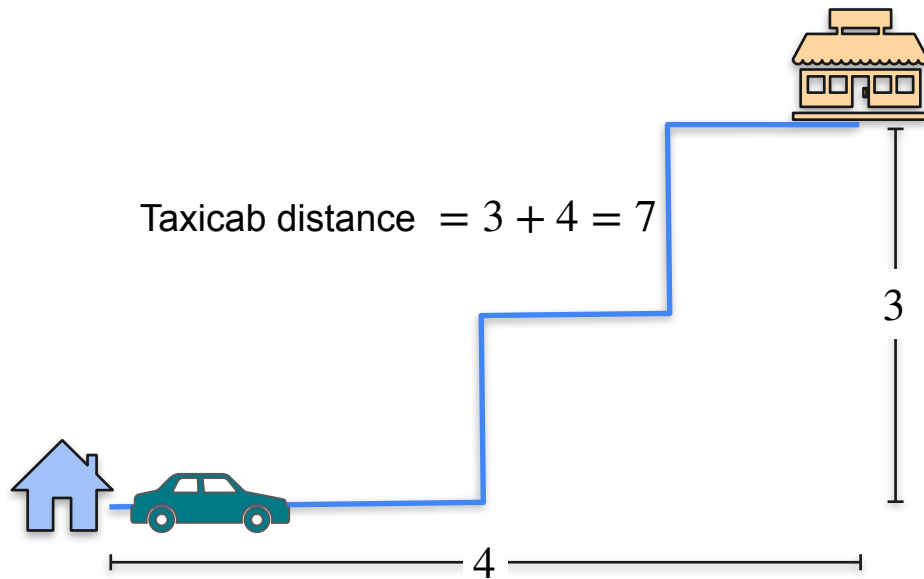
Vectors



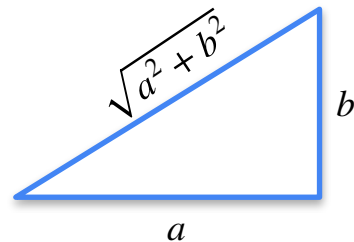
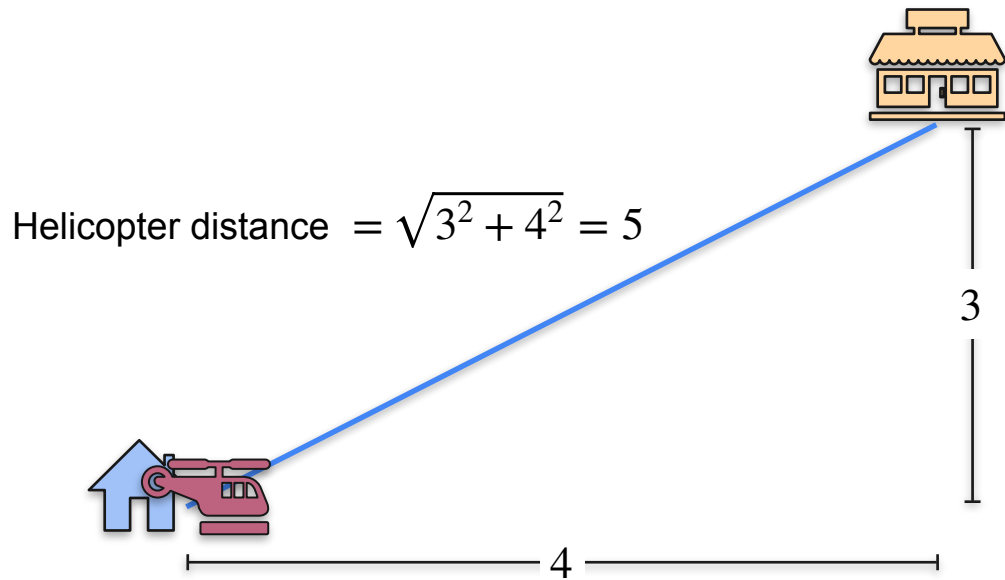
How to get from point A to point B?



How to get from point A to point B?

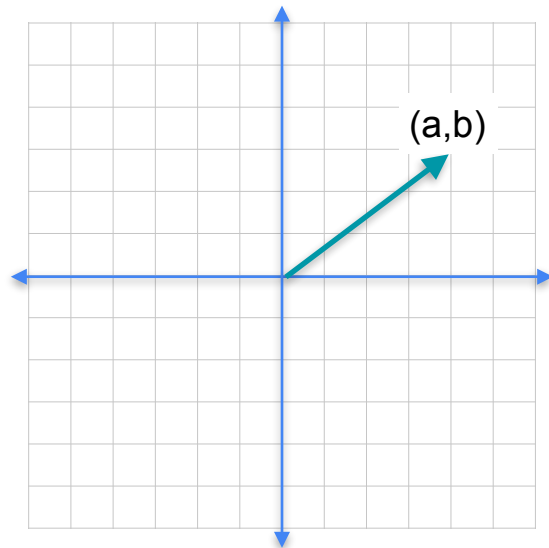


How to get from point A to point B?



Pythagorean Theorem

Norms

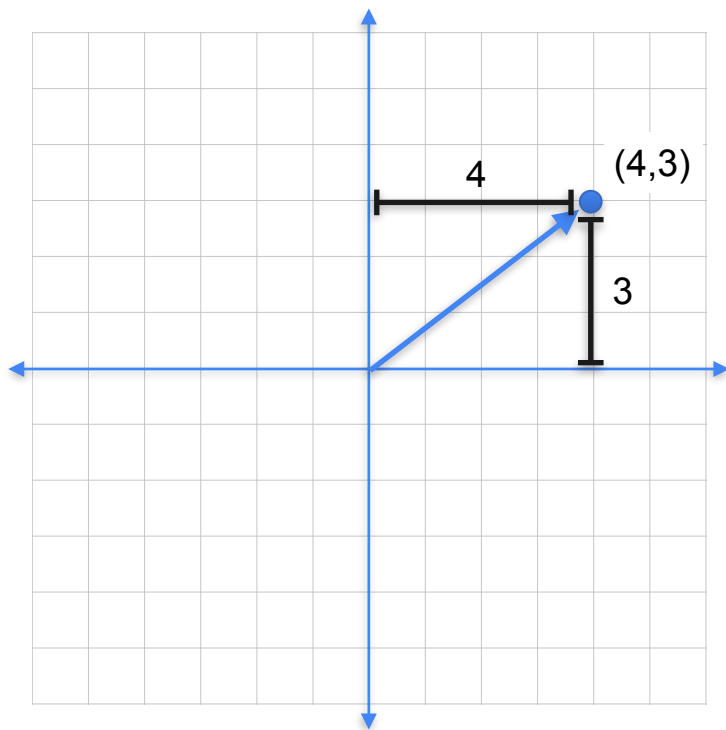


$$\text{L1-norm} = \|(a, b)\|_1 = |a| + |b|$$



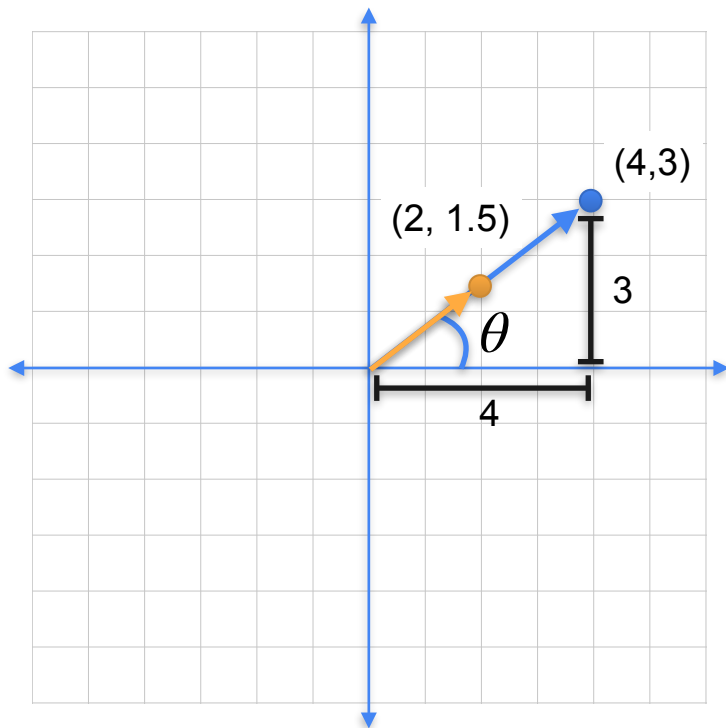
$$\text{L2-norm} = \|(a, b)\|_2 = \sqrt{a^2 + b^2}$$

Norm of a vector



$$\sqrt{4^2 + 3^2} = \sqrt{25} = 5$$

Direction of a vector



$$\tan(\theta) = \frac{3}{4}$$

$$\theta = \arctan(3/4) = 0.64 = 36.87^\circ$$

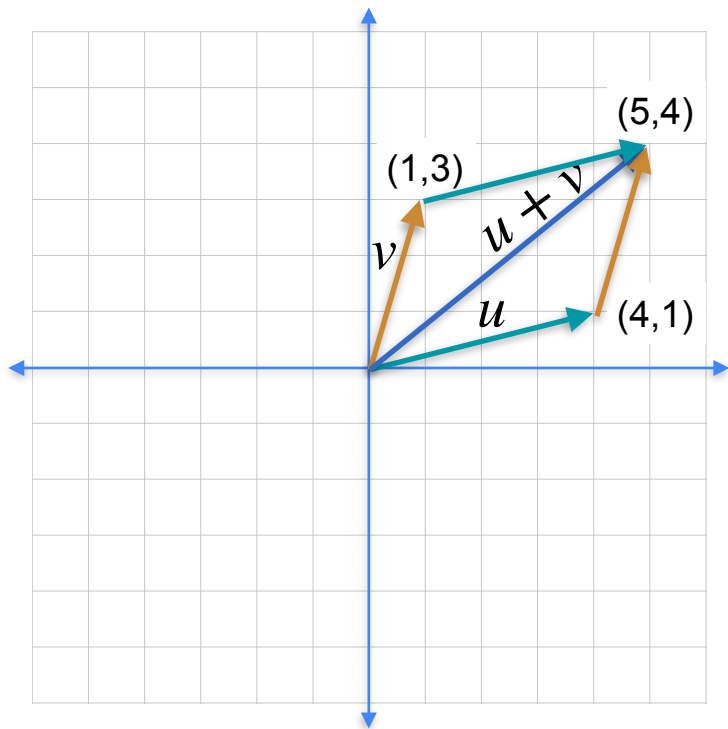


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Vectors and Linear Transformations

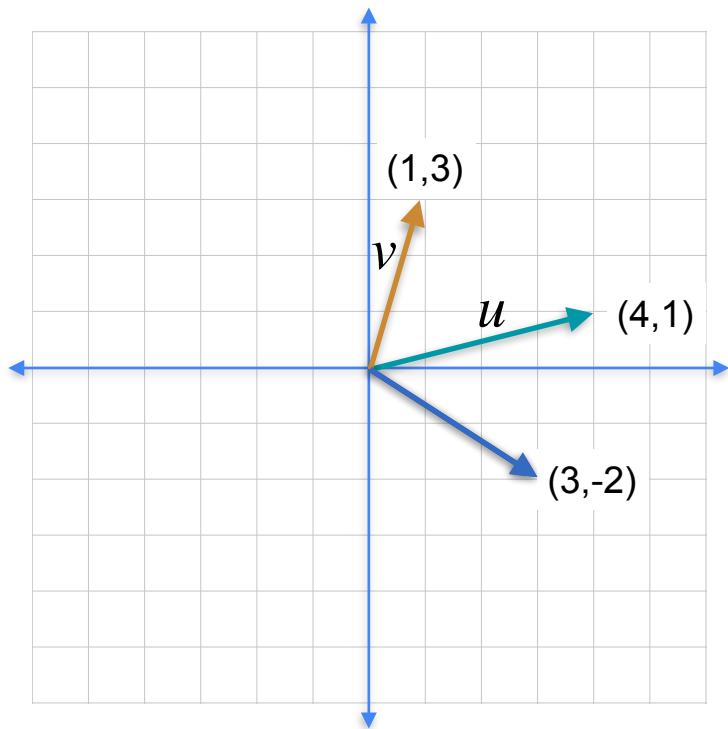
Sum and difference of vectors

Sum of vectors



$$u + v = (4 + 1, 1 + 3) = (5, 4)$$

Difference of vectors



$$u - v = (4 - 1, 1 - 3) = (3, -2)$$

General definition: sum and difference

Same number
of components


$$x = (x_1 \ x_2 \ \dots \ x_n) \qquad y = (y_1 \ y_2 \ \dots \ y_n)$$

Sum

$$x + y = (x_1 + y_1 \ x_2 + y_2 \ \dots \ x_n + y_n)$$

Sum component by component

Difference

$$x - y = (x_1 - y_1 \ x_2 - y_2 \ \dots \ x_n - y_n)$$

Subtract component by component

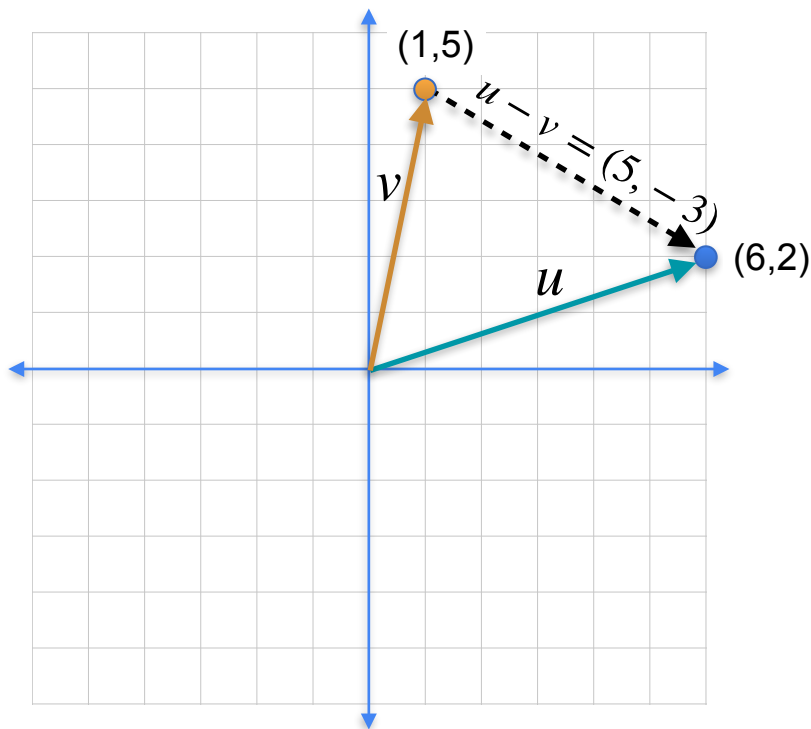


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Vectors and Linear Transformations

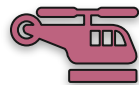
Distance between vectors

Distances



L1-distance

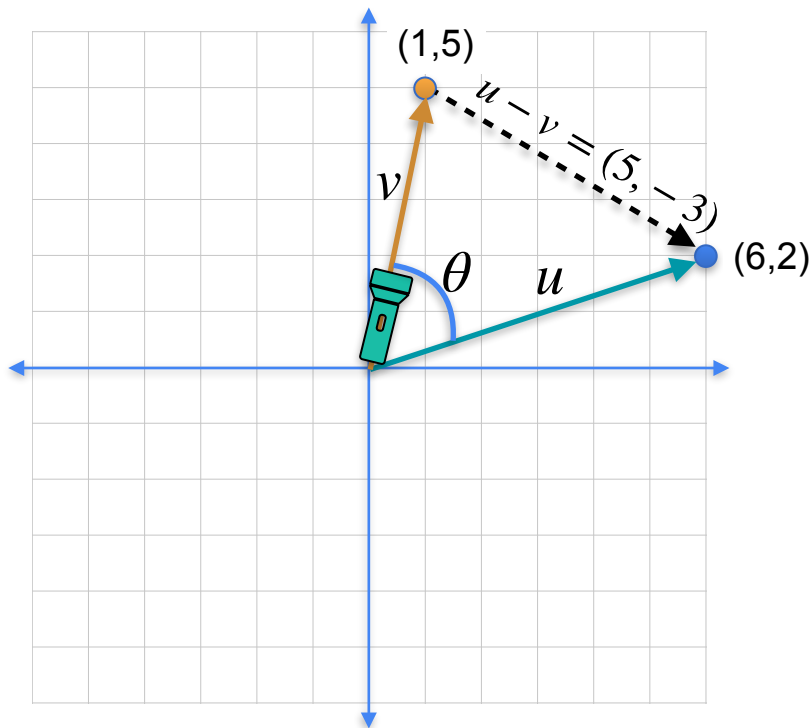
$$\|u - v\|_1 = |5| + |-3| = 8$$



L2-distance

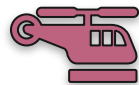
$$\|u - v\|_2 = \sqrt{5^2 + 3^2} = 5.83$$

Distances



L1-distance

$$\|u - v\|_1 = |5| + |-3| = 8$$



L2-distance

$$\|u - v\|_2 = \sqrt{5^2 + 3^2} = 5.83$$



Cosine distance

$$\cos(\theta)$$

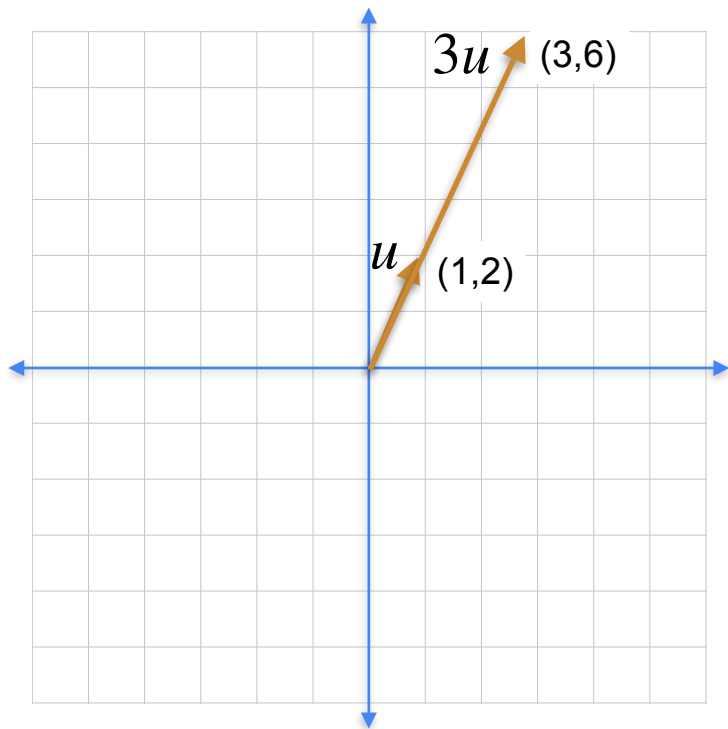


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Vectors and Linear Transformations

Multiplying a vector by a scalar

Multiplying a vector by a scalar

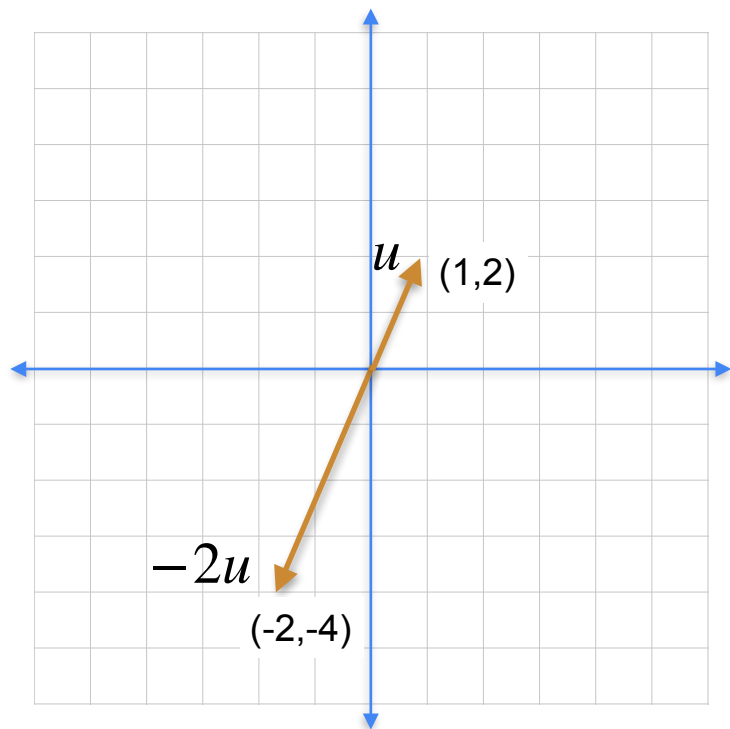


$$u = (1,2)$$

$$\lambda = 3$$

$$\lambda u = (3,6)$$

If the scalar is negative



$$u = (1,2)$$

$$\lambda = -2$$

$$\lambda u = (-2, -4)$$

General definition

Multiplication by a Scalar

$$x = (x_1 \quad x_2 \quad \dots \quad x_n)$$

$$\lambda x = (\lambda x_1 \quad \lambda x_2 \quad \dots \quad \lambda x_n)$$

Video 6: The dot product



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Vectors and Linear Transformations

The dot product

A shortcut for linear operations

Quantities

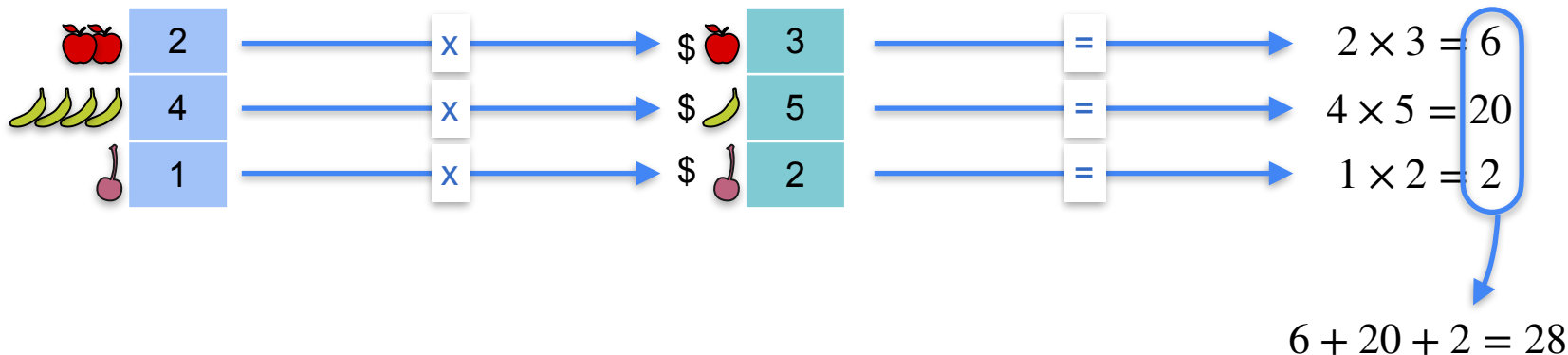
2 apples
4 bananas
1 cherry

Prices

apples: \$3
bananas: \$5
cherries: \$2

Total price

\$28



The dot product

The diagram illustrates the dot product of two vectors using fruit prices. The first vector (blue boxes) represents quantities: 2 apples, 4 bananas, and 1 cherry. The second vector (teal boxes) represents prices: \$3 for an apple, \$5 for a banana, and \$2 for a cherry. The dot product is calculated as the sum of the products of corresponding elements: $2 \times \$3 + 4 \times \$5 + 1 \times \$2 = \28 .

	2
	4
	1

·

\$ 	3
\$ 	5
\$ 	2

=

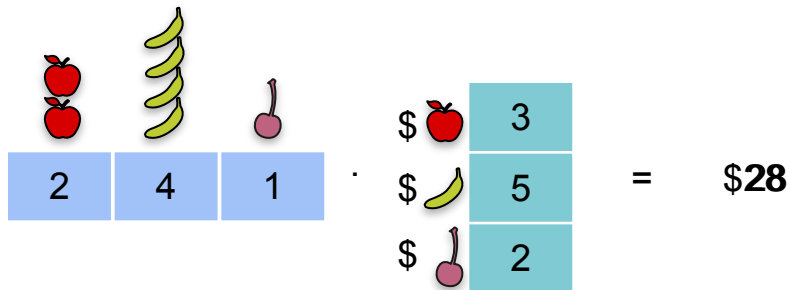
\$28

The dot product

	2	·	\$ 	3	=	\$28
	4		\$ 	5		
	1		\$ 	2		

$$(2 \times 3) + (4 \times 5) + (1 \times 2) = 28$$

The dot product



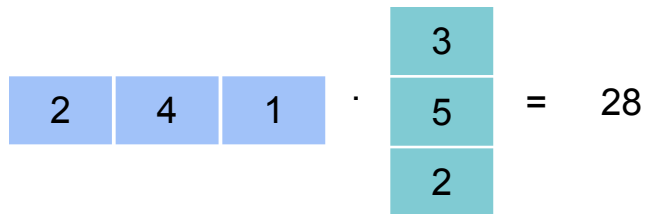
The diagram illustrates a dot product calculation for fruit prices. On the left, a vector of quantities is shown with blue boxes containing the numbers 2, 4, and 1. Above these boxes are icons: two red apples above the 2, four yellow bananas above the 4, and one red cherry above the 1. This vector is multiplied (indicated by a dot) by a column vector of prices. The price vector is shown as a teal column with values 3, 5, and 2. To the left of these values are dollar signs and fruit icons: a red apple next to 3, a yellow banana next to 5, and a red cherry next to 2. The result of the dot product is shown as **\$28**.

2	4	1
\$ 🍎 3	\$ 🍌 5	\$ 🍒 2

= **\$28**

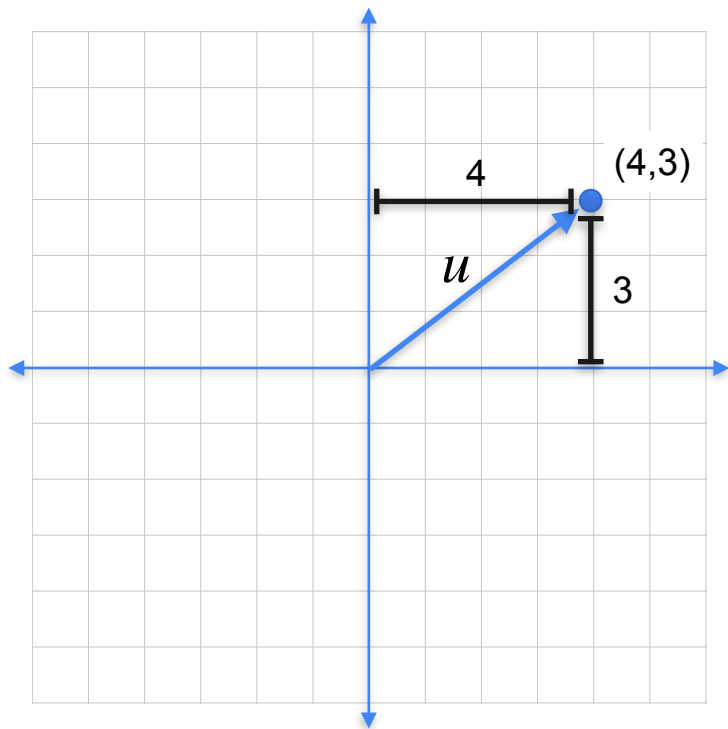
$$(2 \times 3) + (4 \times 5) + (1 \times 2) = 28$$

The dot product


$$\begin{bmatrix} 2 & 4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix} = 28$$

$$(2 \times 3) + (4 \times 5) + (1 \times 2) = 28$$

Norm of a vector using dot product



$$\sqrt{4^2 + 3^2} = \sqrt{25} = 5$$

$$\begin{bmatrix} 4 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \end{bmatrix} = 25$$

$$L2 - norm = \sqrt{\text{dot product}(u, u)}$$

$$\|u\|_2 = \sqrt{\langle u, u \rangle}$$

Vector Transpose

$$\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix} = 28$$

$$(2 \times 3) + (4 \times 5) + (1 \times 2) = 28$$

Vector Transpose

Transpose: convert columns to rows

$$\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix} = 28$$

$$2 \cdot 3 + 4 \cdot 5 + 1 \cdot 2 = 28$$

$$\begin{bmatrix} 2 & 4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 5 \\ 2 \end{bmatrix} = 28$$

$$2 \cdot 3 + 4 \cdot 5 + 1 \cdot 2 = 28$$

Vector Transpose

$$\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}^T =$$

Vector Transpose

$$\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}^T = \begin{bmatrix} 2 & 4 & 1 \end{bmatrix}$$

Vector Transpose

$$\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}^T = \begin{bmatrix} 2 & 4 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 1 \end{bmatrix}^T = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

Vector Transpose

$$\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}^T = \begin{bmatrix} 2 & 4 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 4 & 1 \end{bmatrix}^T = \begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix}$$

Matrix Transpose

2	5
4	7
1	3

$$^T =$$

Transpose the columns!

Matrix Transpose

$$\begin{bmatrix} 2 & 5 \\ 4 & 7 \\ 1 & 3 \end{bmatrix}^T = \begin{bmatrix} 2 & 4 & 1 \end{bmatrix}$$

Matrix Transpose

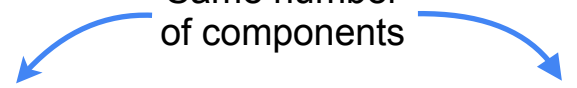
$$\begin{array}{|c|c|} \hline 2 & 5 \\ \hline 4 & 7 \\ \hline 1 & 3 \\ \hline \end{array} \overset{T}{=} \begin{array}{|c|c|c|} \hline 2 & 4 & 1 \\ \hline 5 & 7 & 3 \\ \hline \end{array}$$

3×2 3×2

Columns $\rightarrow$ Rows

General definition: dot product

Same number
of components


$$x = (x_1 \ x_2 \ \dots \ x_n) \qquad y = (y_1 \ y_2 \ \dots \ y_n)$$

$$x \cdot y = (x_1 \times y_1) + (x_2 \times y_2) + \dots + (x_n \times y_n)$$

$\langle x, y \rangle$ is another notation for the dot product

$$x \cdot y^T = (x_1 \ x_2 \ \dots \ x_n) \cdot \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix}$$

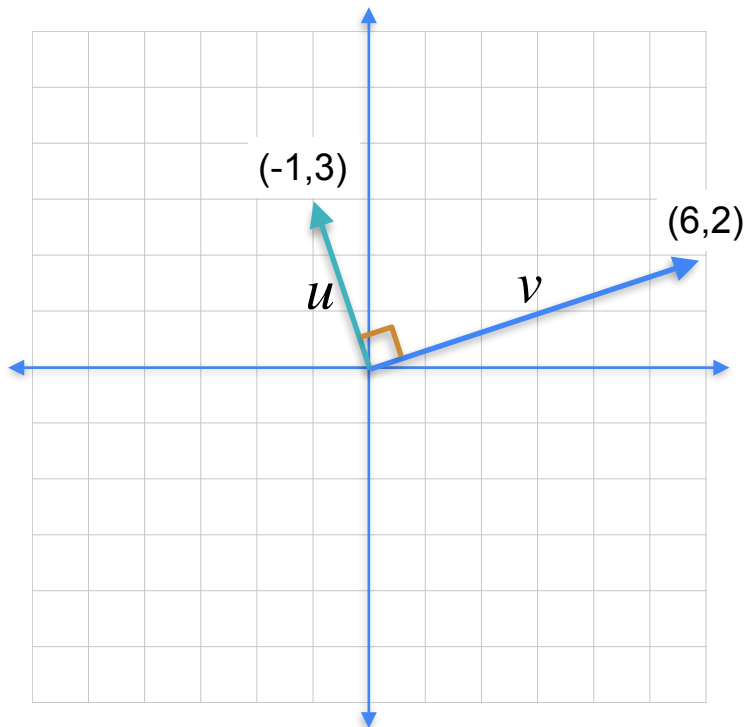


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Vectors and Linear Transformations

Geometric dot product

Orthogonal vectors have dot product 0



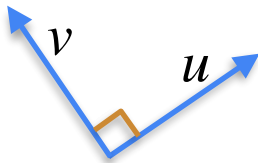
$$\begin{bmatrix} 6 & 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 3 \end{bmatrix} = 0$$

$$\langle u, v \rangle = 0$$

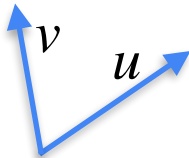
The dot product



$$\langle u, u \rangle = \|u\|^2$$

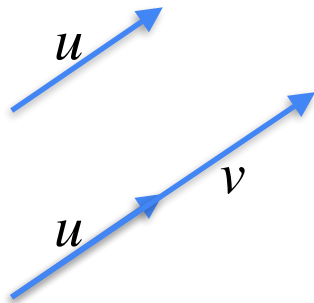


$$\langle u, v \rangle = 0$$



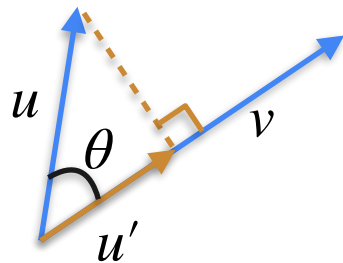
$$\langle u, v \rangle = ?$$

The dot product



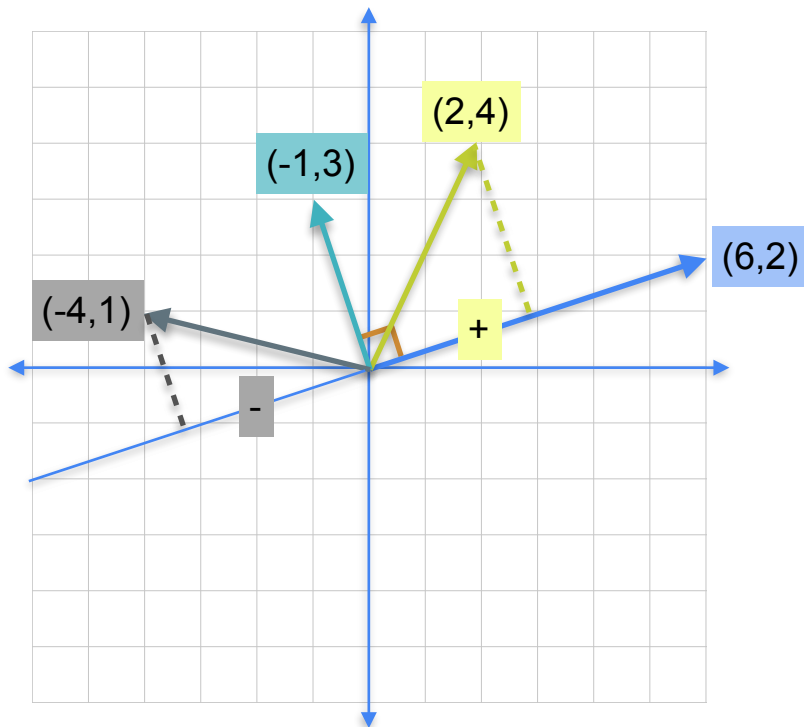
$$\langle u, u \rangle = \|u\|^2 = \|u\| \cdot \|u\|$$

$$\langle u, v \rangle = \|u\| \cdot \|v\|$$



$$\begin{aligned}\langle u, v \rangle &= \|u'\| \cdot \|v\| \\ &= \|u\| \|v\| \cos(\theta)\end{aligned}$$

Geometric dot product



$$\begin{bmatrix} 6 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 2 \\ 4 \end{bmatrix} = 20 \quad \text{Positive}$$

$$\begin{bmatrix} 6 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 3 \end{bmatrix} = 0$$

$$\begin{bmatrix} 6 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} -4 \\ 1 \end{bmatrix} = -22 \quad \text{Negative}$$

Geometric dot product

Dot product with $u > 0$

$$\langle u, v \rangle > 0$$

$$\langle u, v \rangle = 0$$

$$\langle u, v \rangle < 0$$

Dot product with $u < 0$

Dot product with $u = 0$



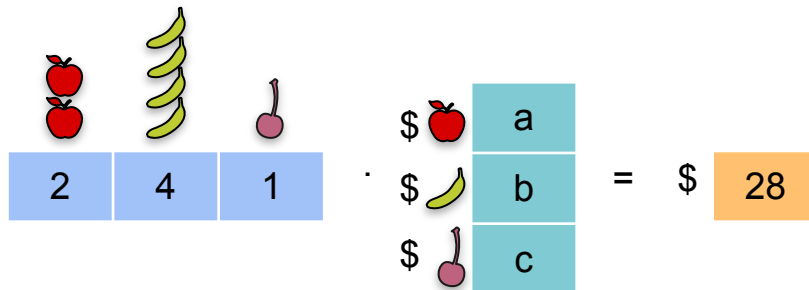
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Vectors and Linear Transformations

**Multiplying a matrix by a
vector**

Equations as dot product

$$2a + 4b + c = 28$$



Equations as dot product

$$a + b + c = 10$$

$\begin{bmatrix} 1 & 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} \$ \text{apple} & a \\ \$ \text{banana} & b \\ \$ \text{cherry} & c \end{bmatrix} = \$ 10$

$$a + 2b + c = 15$$

$\begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} \$ \text{apple} & a \\ \$ \text{banana} & b \\ \$ \text{cherry} & c \end{bmatrix} = \$ 15$

$$a + b + 2c = 12$$

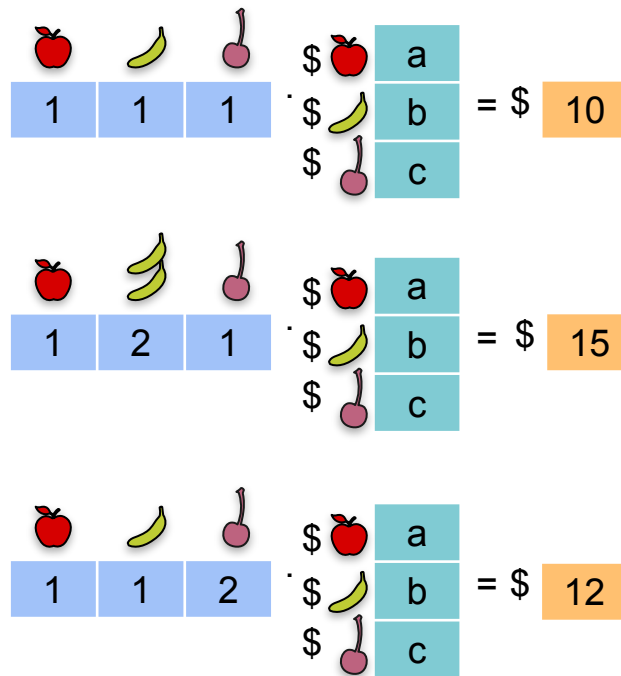
$\begin{bmatrix} 1 & 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} \$ \text{apple} & a \\ \$ \text{banana} & b \\ \$ \text{cherry} & c \end{bmatrix} = \$ 12$

Equations as dot product

$$a + b + c = 10$$

$$a + 2b + c = 15$$

$$a + b + 2c = 12$$

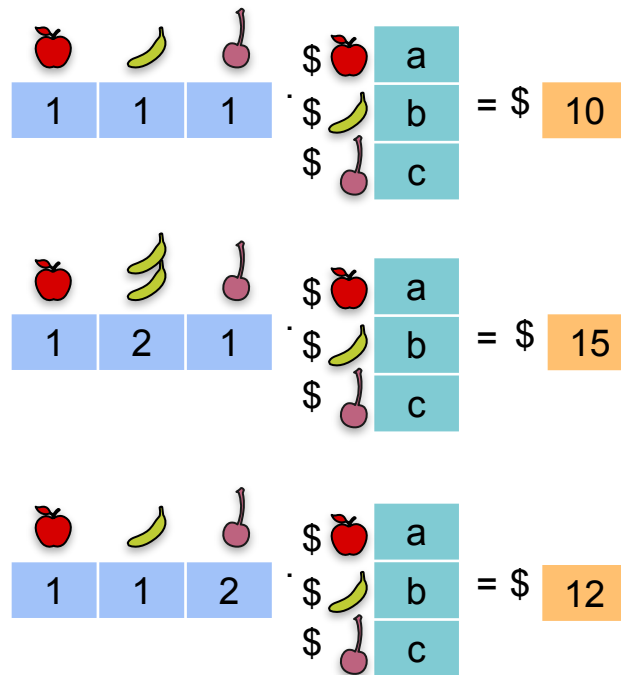


Equations as dot product

$$a + b + c = 10$$

$$a + 2b + c = 15$$

$$a + b + 2c = 12$$



Equations as dot product

System of equations

$$a + b + c = 10$$

$$a + 2b + c = 15$$

$$a + b + 2c = 12$$

Matrix product

		
1	1	1
1	2	1
1	1	2

 ·

\$ 	a
\$ 	b
\$ 	c

 = \$

10
15
12

Equations as dot product

System of equations

$$a + b + c = 10$$

$$a + 2b + c = 15$$

$$a + b + 2c = 12$$

Matrix product

1	1	1	a	=	10
1	2	1	b		15
1	1	2	c		12

Equations as dot product

Matrix product

1	1	1	a
1	2	1	b
1	1	2	c

 =

10
15
12

3×3 Length 3

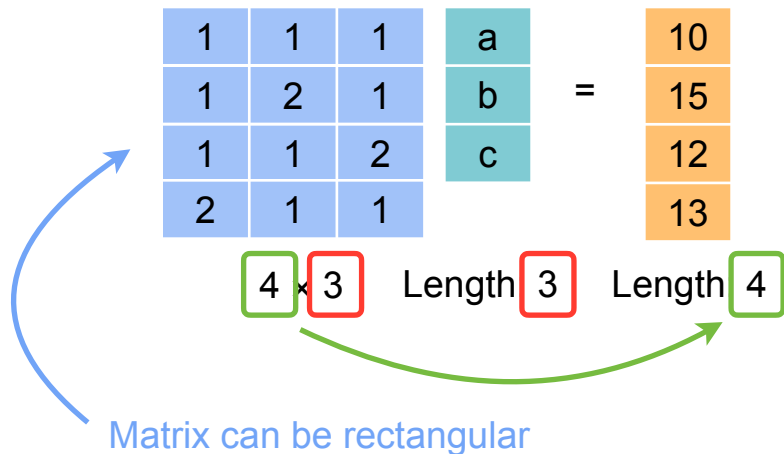
Matrix can be rectangular



columns = length of vector

Equations as dot product

Matrix product



columns = length of vector

W3 Lesson 2

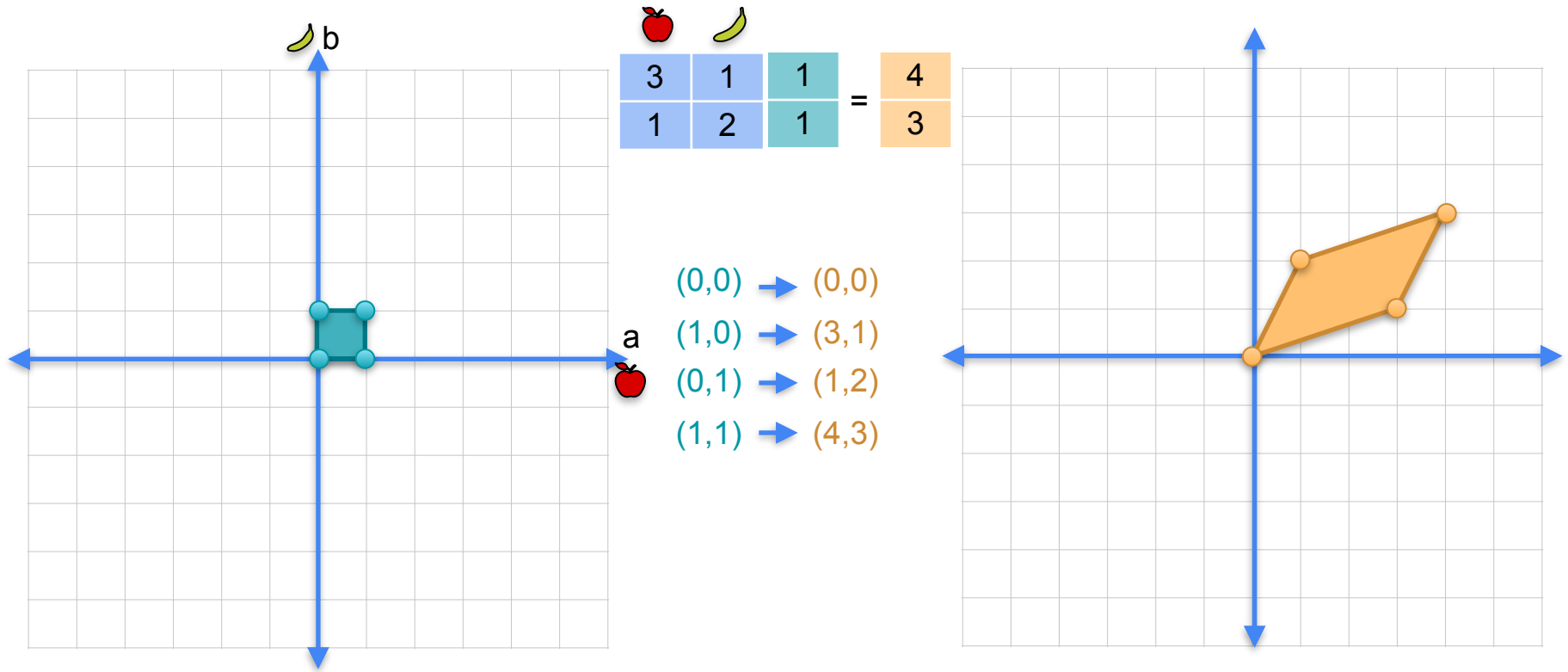


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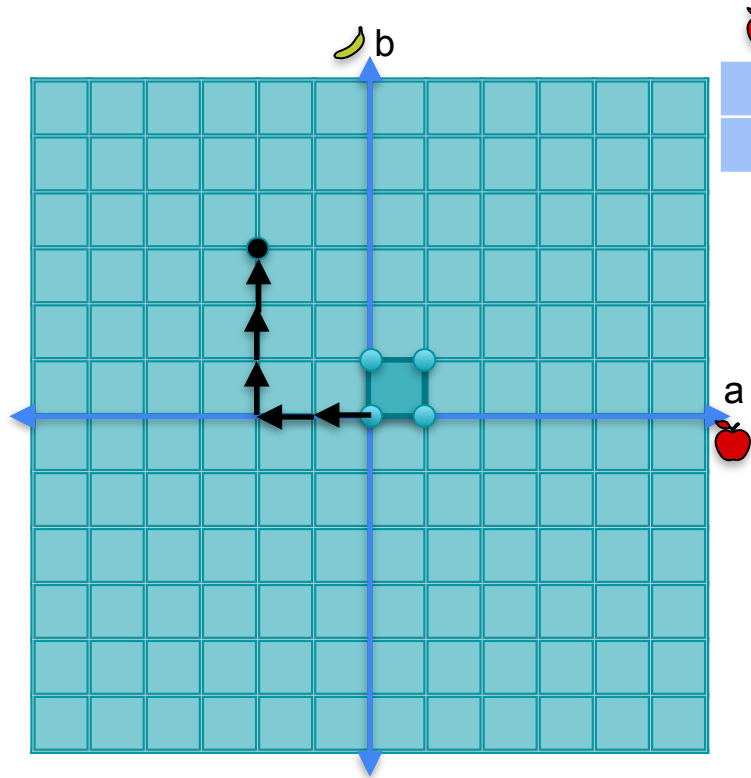
Vectors and Linear Transformations

**Matrices as linear
transformations**

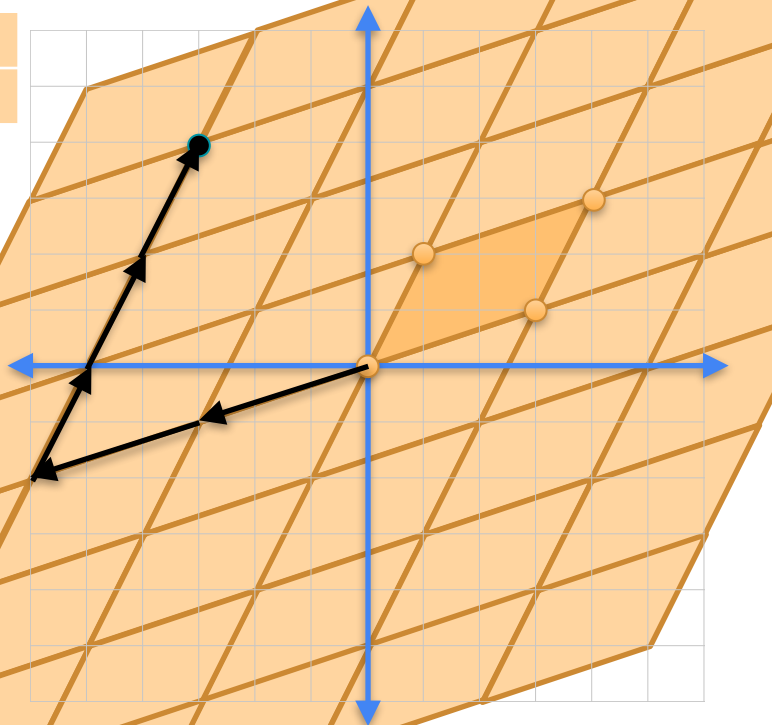
Matrices as linear transformations



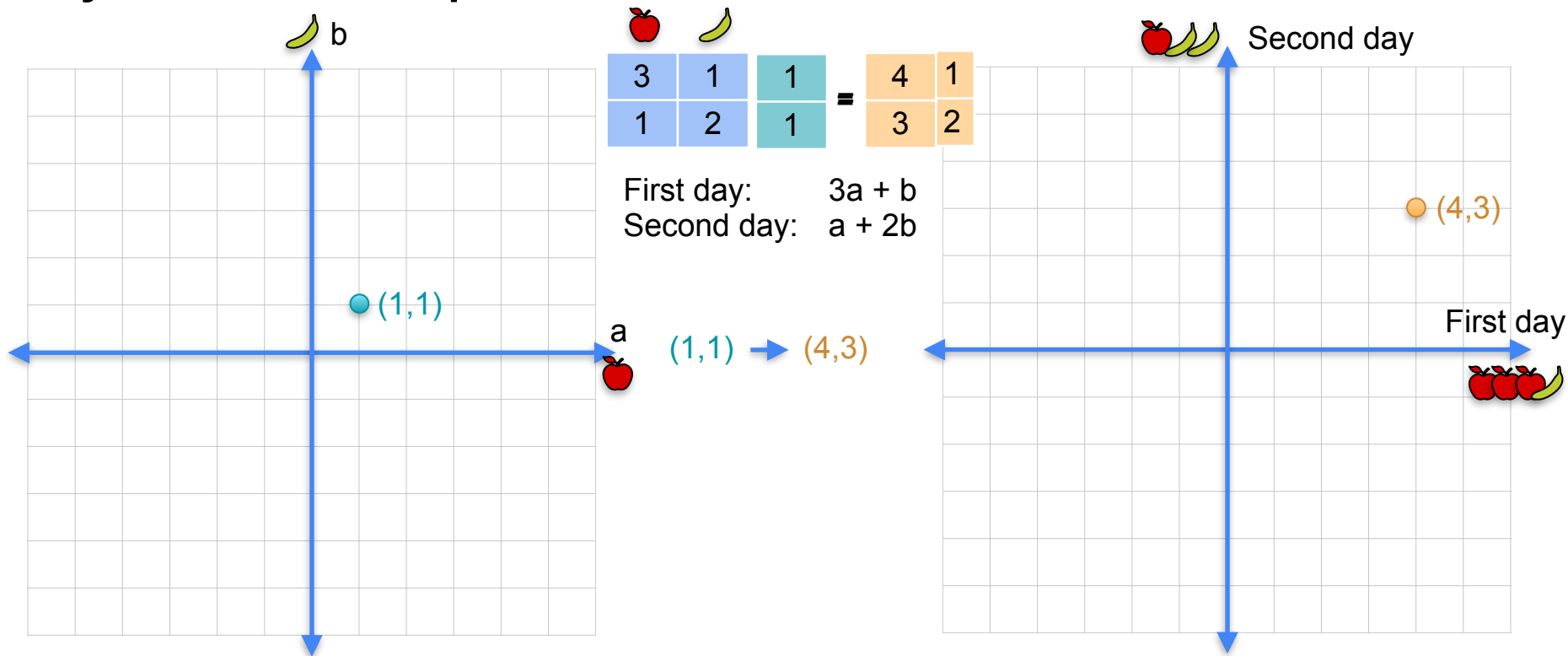
Matrices as linear transformations



$$\begin{matrix} \text{apple} & \text{banana} \\ \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} & \begin{bmatrix} -2 \\ 3 \end{bmatrix} \end{matrix} = \begin{bmatrix} -3 \\ 4 \end{bmatrix}$$



Systems of equations as linear transformations



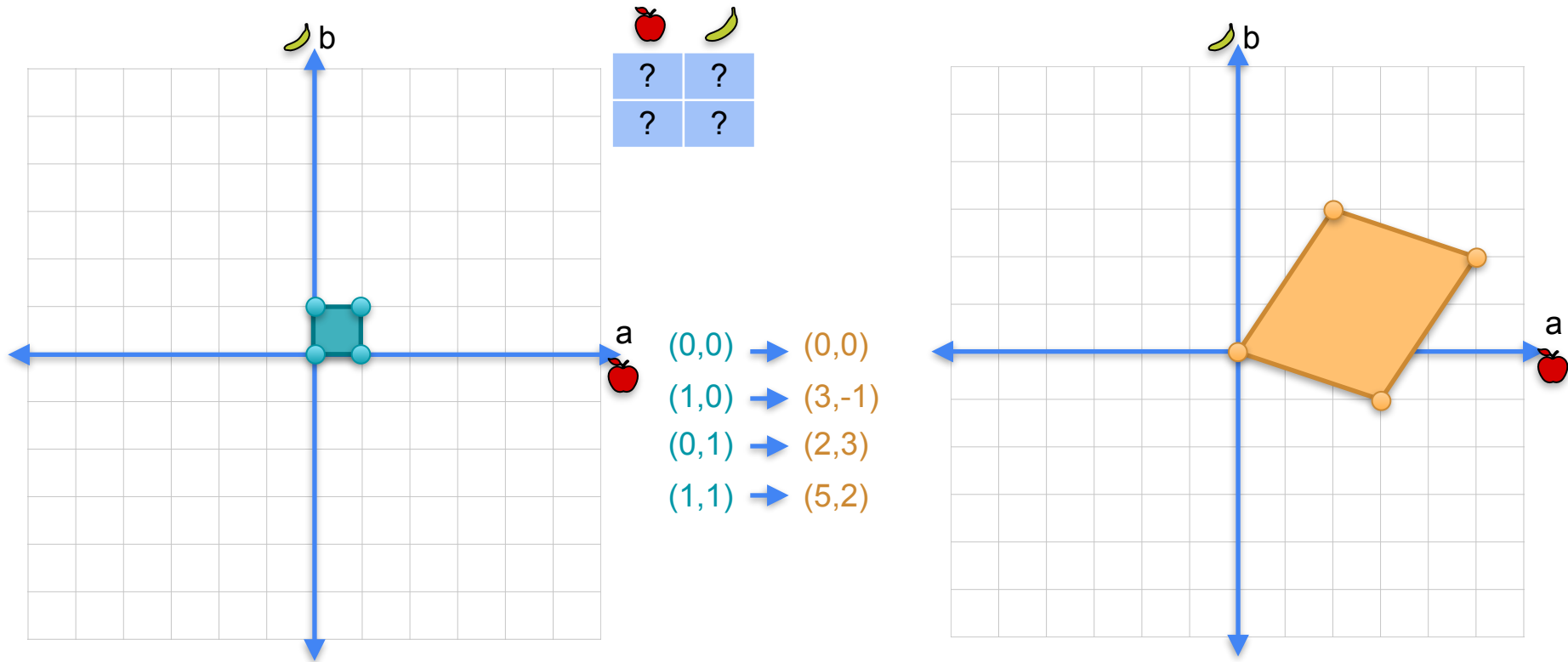


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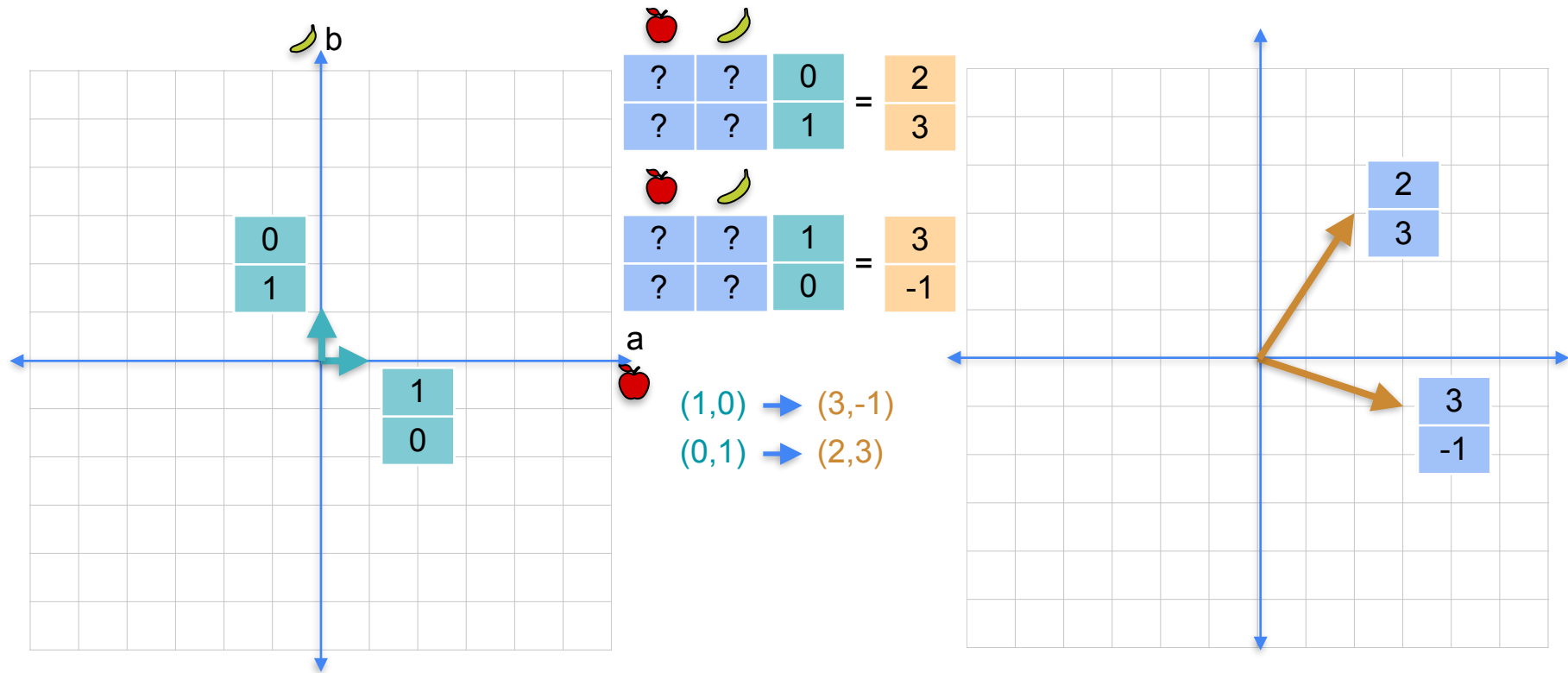
Vectors and Linear Transformations

**Linear transformations as
matrices**

Linear transformations as matrices



Linear transformations as matrices



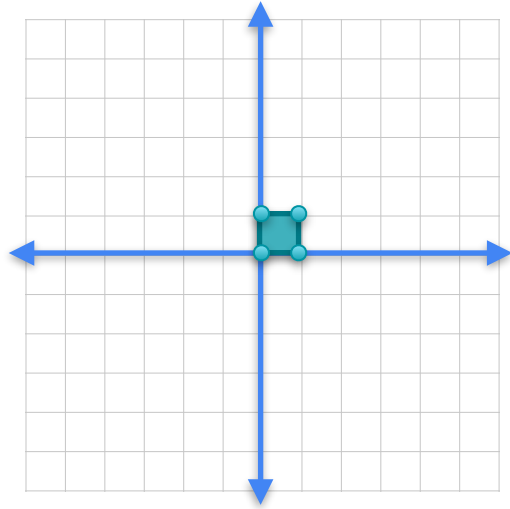


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Vectors and Linear Transformations

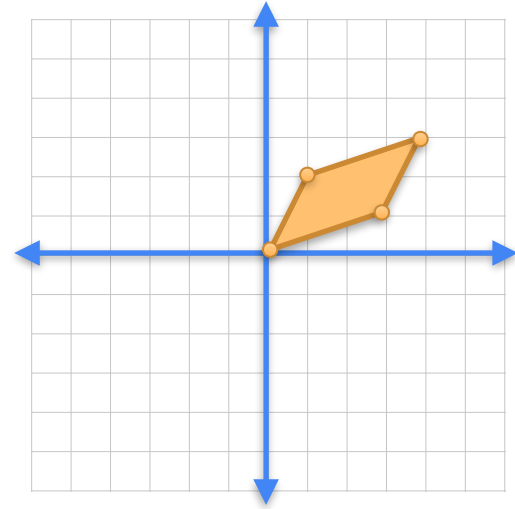
Matrix multiplication

Combining linear transformations

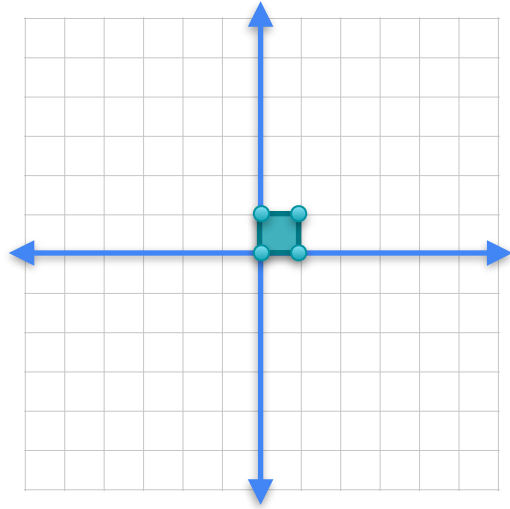


$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

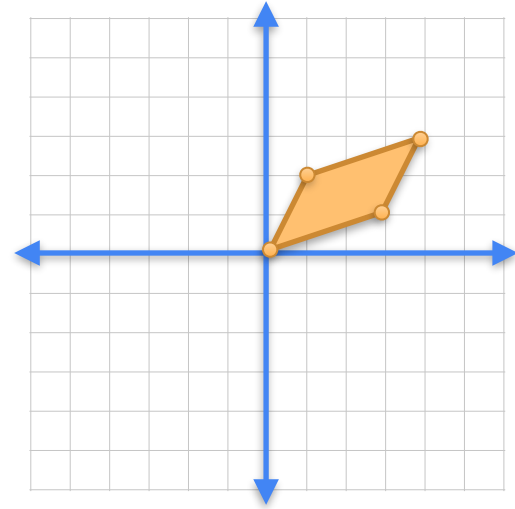


Combining linear transformations

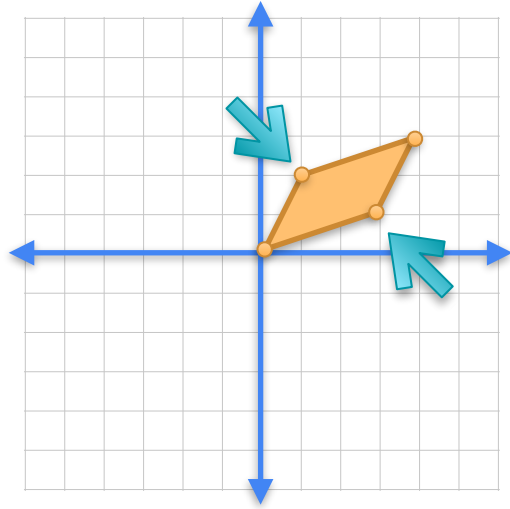


$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

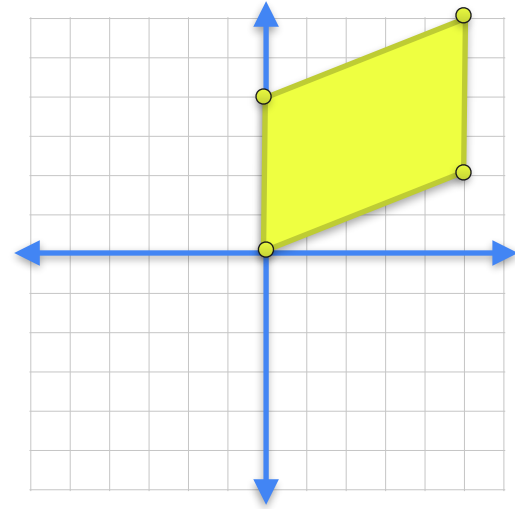


Combining linear transformations

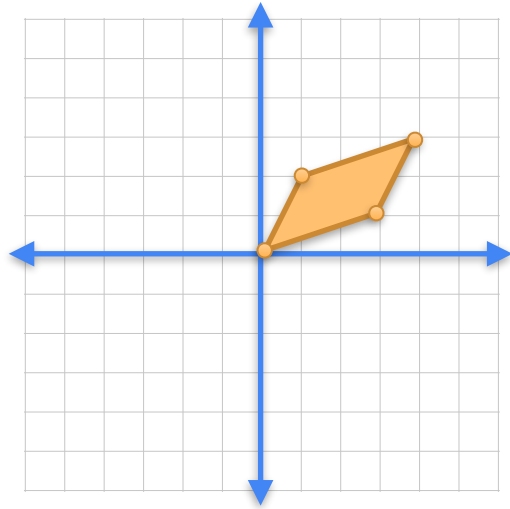


$$\begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

$$\begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$$

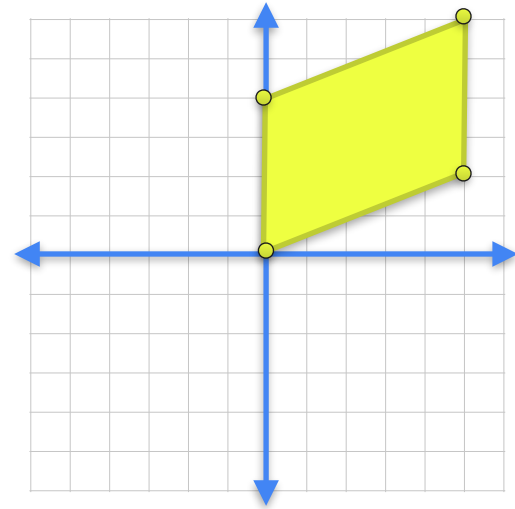


Combining linear transformations

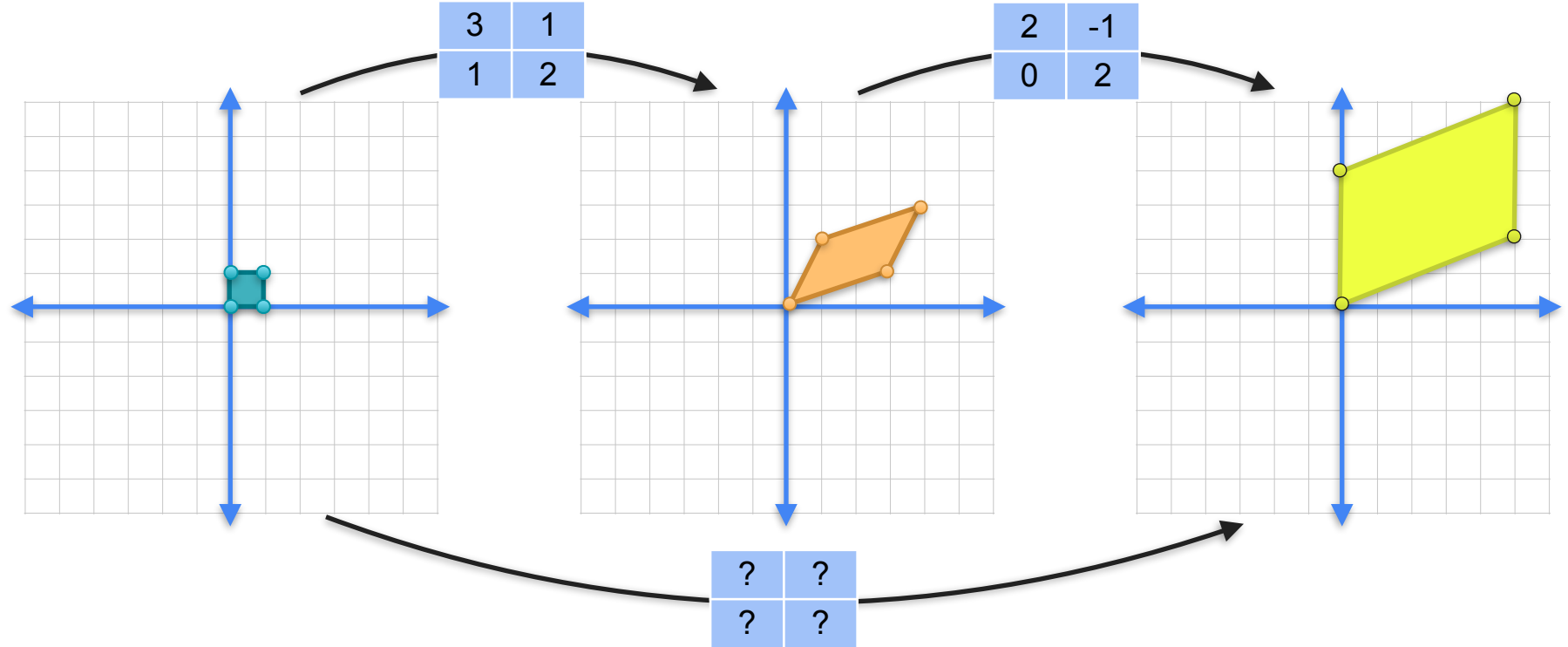


$$\begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

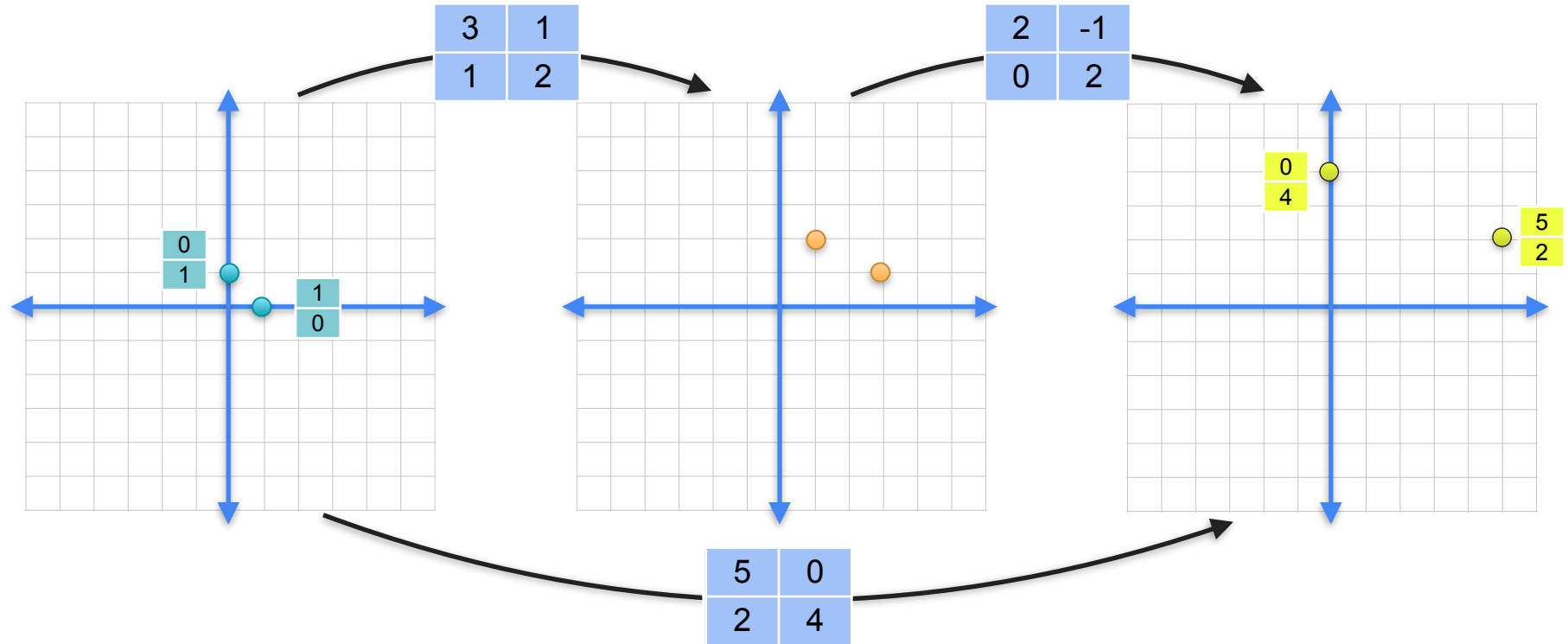
$$\begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 4 \end{bmatrix}$$



Combining linear transformations



Combining linear transformations



Combining linear transformations

$$\begin{array}{c} \text{Second} \\ \downarrow \\ \begin{array}{|c|c|} \hline 2 & -1 \\ \hline 0 & 2 \\ \hline \end{array} \end{array} \cdot \begin{array}{c} \text{First} \\ \downarrow \\ \begin{array}{|c|c|} \hline 3 & 1 \\ \hline 1 & 2 \\ \hline \end{array} \end{array} = \begin{array}{|c|c|} \hline 5 & 0 \\ \hline 2 & 4 \\ \hline \end{array}$$

Dimension of the matrices

The diagram illustrates the dot product of two 2x2 matrices. The first matrix (teal) has elements $\begin{bmatrix} 2 & -1 \\ 0 & 2 \end{bmatrix}$. The second matrix (orange) has elements $\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$. The result is shown as a 2x2 grid of four products, each calculated as the dot product of a row from the first matrix and a column from the second matrix.

Top-left product: $2 \cdot 3 + (-1) \cdot 1 = 6 - 1 = 5$

Top-right product: $2 \cdot 1 + (-1) \cdot 2 = 2 - 2 = 0$

Bottom-left product: $0 \cdot 3 + 2 \cdot 1 = 0 + 2 = 2$

Bottom-right product: $0 \cdot 1 + 2 \cdot 2 = 0 + 4 = 4$

Dimension of the matrices

$$\begin{array}{|c|c|} \hline 2 & -1 \\ \hline 0 & 2 \\ \hline \end{array} \cdot \begin{array}{|c|c|} \hline 3 & 1 \\ \hline 1 & 2 \\ \hline \end{array} = \begin{array}{|c|c|} \hline 5 & 0 \\ \hline 2 & 4 \\ \hline \end{array}$$

$2 \times 2 \quad 2 \times 2 \quad 2 \times 2$

Dimension of the matrices

3	1	4
2	-1	2

2×3

·

3	0	1	-2
1	5	2	0
-2	1	4	0

3×4

=

?			

2×4

Dimension of the matrices

The diagram illustrates the multiplication of two matrices. The first matrix is a 2x3 matrix with elements $\begin{bmatrix} 3 & 1 & 4 \\ 2 & -1 & 2 \end{bmatrix}$. Its dimensions are indicated by a blue box around the number 2 and a red box around the number 3, with the text 2×3 below it. The second matrix is a 3x4 matrix with elements $\begin{bmatrix} 3 & 0 & 1 & -2 \\ 1 & 5 & 2 & 0 \\ -2 & 1 & 4 & 0 \end{bmatrix}$. Its dimensions are indicated by a red box around the number 3 and a purple box around the number 4, with the text 3×4 below it. A red box highlights the third column of the first matrix (elements 4, 2) and the first row of the second matrix (elements 3, 0, 1, -2), showing their match. An equals sign follows, leading to a 2x4 result matrix with elements $\begin{bmatrix} 2 & 9 & 21 & -6 \\ ? & -3 & 8 & -4 \end{bmatrix}$. The dimensions of the result are indicated by a blue box around the number 2 and a purple box around the number 4, with the text 2×4 below it.

2	9	21	-6
?	-3	8	-4

- Columns of first matrix must match rows of second (numbers in red match)
- Result takes number of rows from first matrix (numbers in blue match)
- Result takes number of columns from second matrix (numbers in purple match)

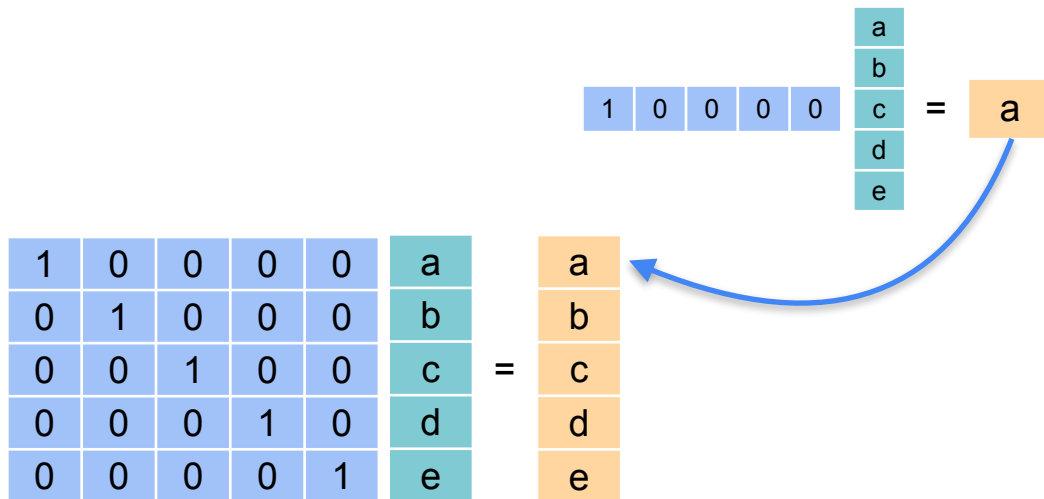


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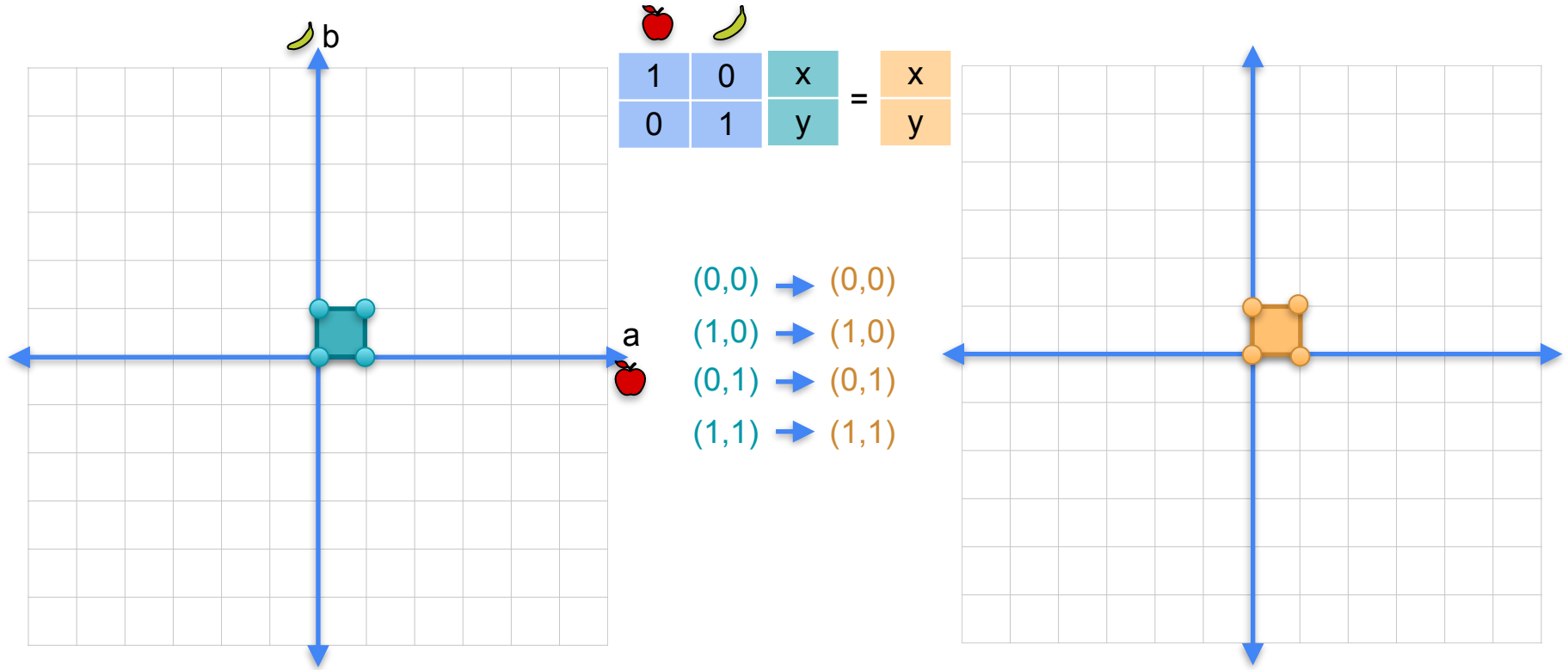
Vectors and Linear Transformations

The identity matrix

The identity matrix



The identity matrix



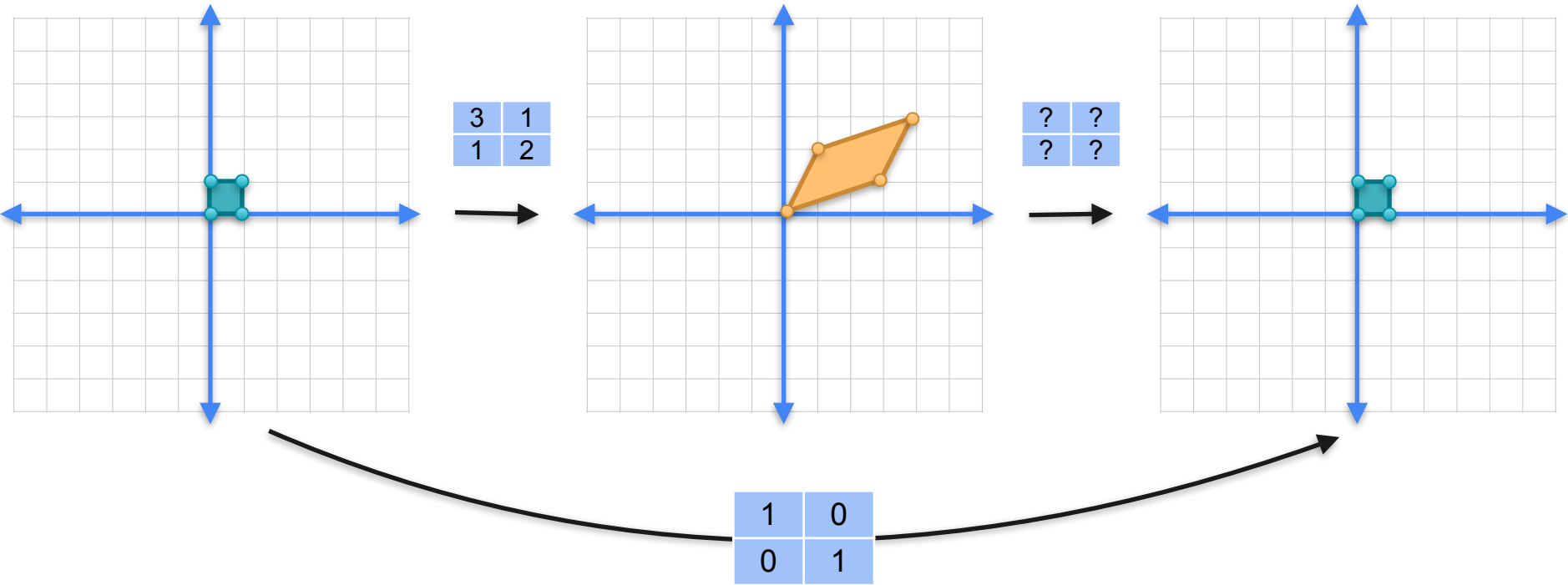


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Vectors and Linear Transformations

Matrix inverse

Matrix inverses



Multiplying matrices

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$



$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}^{-1} = \begin{bmatrix} 2/5 & -1/5 \\ -1/5 & 3/5 \end{bmatrix}$$

How to find an inverse?

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} a \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$3a + 1c = 1$$

$$a = \frac{2}{5}$$

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$3b + 1d = 0$$

$$b = -\frac{1}{5}$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} a \\ c \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$1a + 2c = 0$$

$$c = -\frac{1}{5}$$

$$\begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} b \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$1b + 2d = 1$$

$$d = \frac{3}{5}$$

Quiz

- Find the inverse of the following matrix. If you find that the task is impossible, feel free to click on “I couldn’t find it”

5	2
1	2

Solution

- By solving the corresponding system of linear equations, we get the following.

$$\begin{bmatrix} 5 & 2 \\ 1 & 2 \end{bmatrix} \cdot \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 5 & 2 \end{bmatrix} \begin{bmatrix} a \\ c \end{bmatrix} = 1$$

$$\bullet 5a + 2c = 1$$

$$\bullet a = 1/4$$

$$\begin{bmatrix} 5 & 2 \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix} = 0$$

$$\bullet 5b + 2d = 0$$

$$\bullet b = -1/4$$

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} a \\ c \end{bmatrix} = 0$$

$$\bullet a + 2c = 0$$

$$\bullet c = -1/8$$

$$\begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} b \\ d \end{bmatrix} = 1$$

$$\bullet b + 2d = 1$$

$$\bullet d = 5/8$$

Quiz

- Find the inverse of the following matrix. If you find that the task is impossible, feel free to click on “I’m reaching a dead end”

1	1
2	2

Solutions

- The inverse doesn't exist!

We need to solve the following system of linear equations:

$$\begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$a + c = 1$$

$$2b + 2d = 1$$

$$2a + 2c = 0$$

$$b + d = 0$$

This is clearly a contradiction, since equation 1 says $a+c=1$, and equation 3 says $2a+2c=0$.



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Vectors and Linear Transformations

**Which matrices have an
inverse?**

Which matrices have inverses?

$$5^{-1} = 0.2$$

$$8^{-1} = 0.125$$

$$0^{-1} = ???$$

3	1
1	2

⁻¹ =

0.4	-0.2
-0.2	0.6

Non-singular matrix
Invertible

$$\text{Det} = 5$$

5	2
1	2

⁻¹ =

0.25	-0.25
-0.125	0.625

Non-singular matrix
Invertible

$$\text{Det} = 8$$

Non-zero determinants

1	1
2	2

 =

?	?
?	?

Singular matrix
Non-invertible

$$\text{Det} = 0$$

Zero determinant



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Vectors and Linear Transformations

**Neural networks and
matrices**

Quiz: Natural language processing

Spam	Lottery	Win
Yes	1	1
Yes	2	1
No	0	0
Yes	0	2
No	0	1
No	1	0
Yes	2	2
Yes	2	0
Yes	1	2

Scores:

Lottery: ____ points

Win: ____ points

Examples

Lottery: 3 point

Win: 2 points

“Win, win the lottery!” : 7points

Rule:

If the number of points of the sentence is bigger than ____, then the email is spam.

Goal: Find the best points and threshold

Lottery: ____ point

Win: ____ point

Threshold: ____ points

Quiz: Natural language processing

Spam	Lottery	Win
Yes	1	1
Yes	2	1
No	0	0
Yes	0	2
No	0	1
No	1	0
Yes	2	2
Yes	2	0
Yes	1	2

Score	> 1.5?
2	Yes
3	Yes
0	No
2	Yes
1	No
1	No
4	Yes
2	Yes
3	Yes

Solution:

Lottery: 1 point

Win: 1 point

Threshold: 1.5 points

Graphical natural language processing

Spam	Lottery	Win
Yes	1	1
Yes	2	1
No	0	0
Yes	0	2
No	0	1
No	1	0
Yes	2	2
Yes	2	0
Yes	1	2

Line:

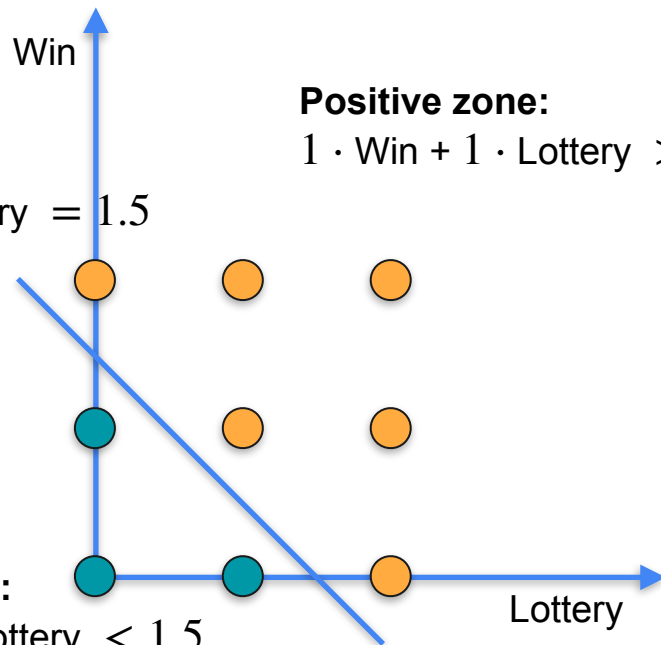
$$1 \cdot \text{Win} + 1 \cdot \text{Lottery} = 1.5$$

Negative zone:

$$1 \cdot \text{Win} + 1 \cdot \text{Lottery} < 1.5$$

Positive zone:

$$1 \cdot \text{Win} + 1 \cdot \text{Lottery} > 1.5$$



Graphical natural language processing

Spam	Lottery	Win
Yes	1	1
Yes	2	1
No	0	0
Yes	0	2
No	0	1
No	1	0
Yes	2	2
Yes	2	0
Yes	1	2

Model
1
1

= 3

Check: > 1.5?



Spam

Dot product between vectors

Spam	Lottery	Win
Yes	1	1
Yes	2	1
No	0	0
Yes	0	2
No	0	1
No	1	0
Yes	2	2
Yes	2	0
Yes	1	2

Model
1
1

= 1

Check: > 1.5?



Not spam

Matrix multiplication

Spam	Lottery	Win
Yes	1	1
Yes	2	1
No	0	0
Yes	0	2
No	0	1
No	1	0
Yes	2	2
Yes	2	0
Yes	1	2

Model
1
1

=

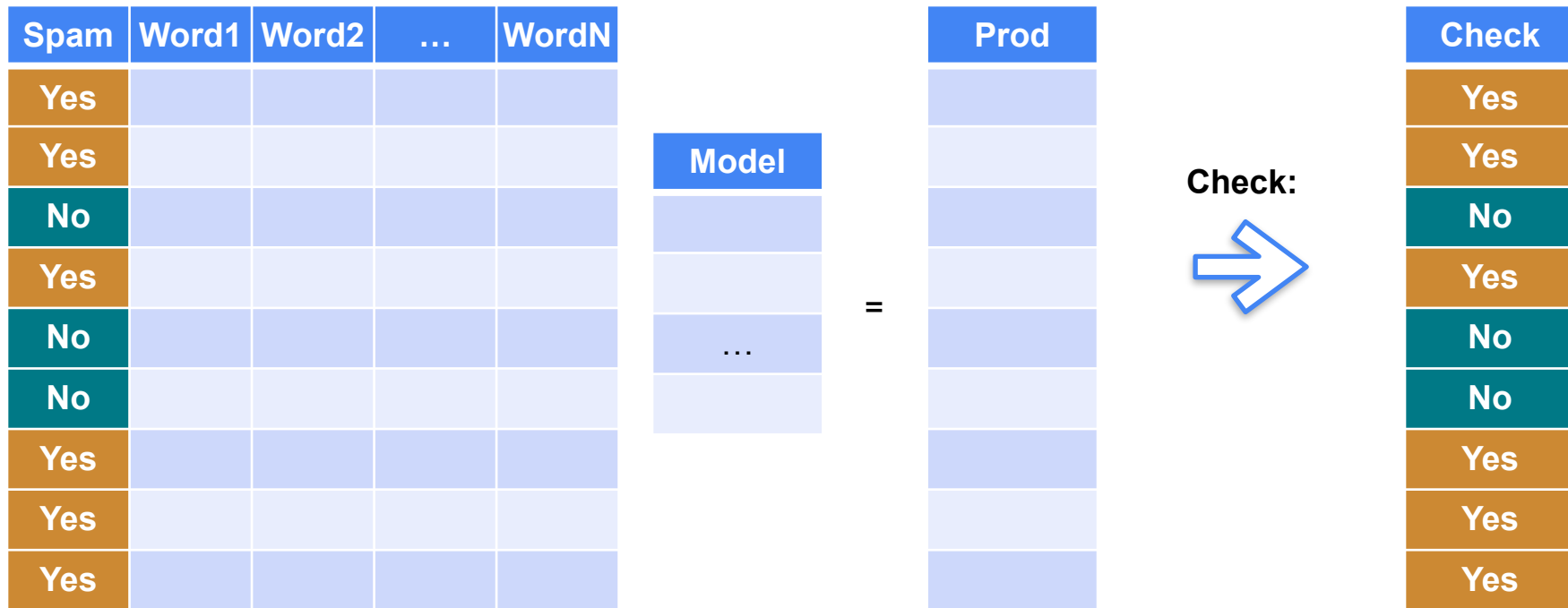
Prod
2
3
0
2
1
1
4
2
3

Check: >1.5?



Check
Yes
Yes
No
Yes
No
No
Yes
Yes
Yes

Perceptrons



Threshold and bias

Spam	Lottery	Win	Bias
Yes	1	1	1
Yes	2	1	1
No	0	0	1
Yes	0	2	1
No	0	1	1
No	1	0	1
Yes	2	2	1
Yes	2	0	1
Yes	1	2	1

Check

$$1 \cdot \text{Win} + 1 \cdot \text{Lottery} - 1.5 > 0$$

Threshold

$$1 \cdot \text{Win} + 1 \cdot \text{Lottery} - 1.5 > 0$$

Bias

Model

1
1
-1.5

Bias

Check: > 0?

The AND operator

AND	x	y
No	0	0
No	1	0
No	0	1
Yes	1	1

Model
1
1

=

Dot prod
0
1
1
2

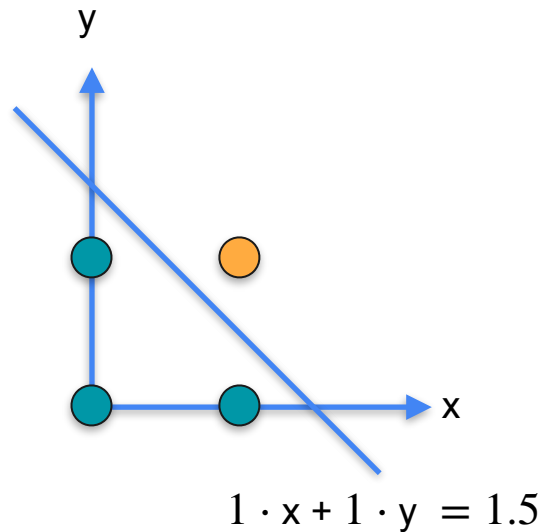
Check: >1.5?



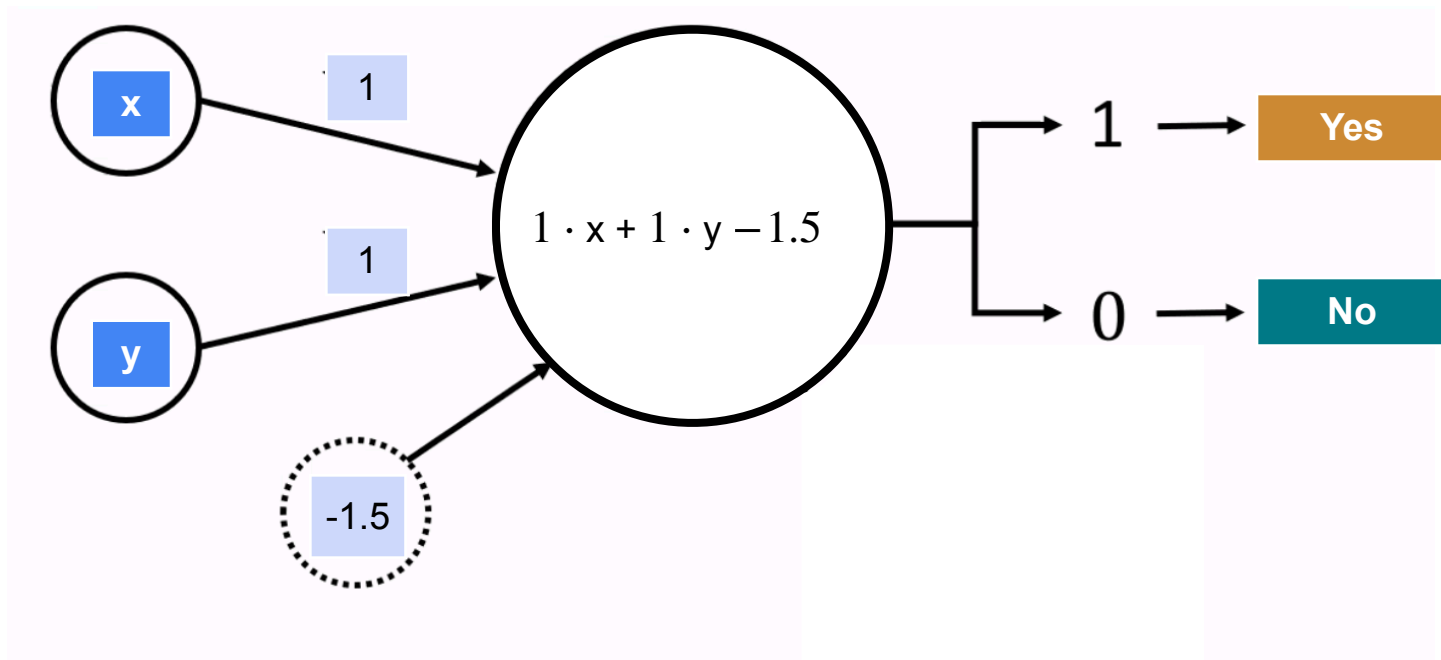
Check
No
No
No
Yes

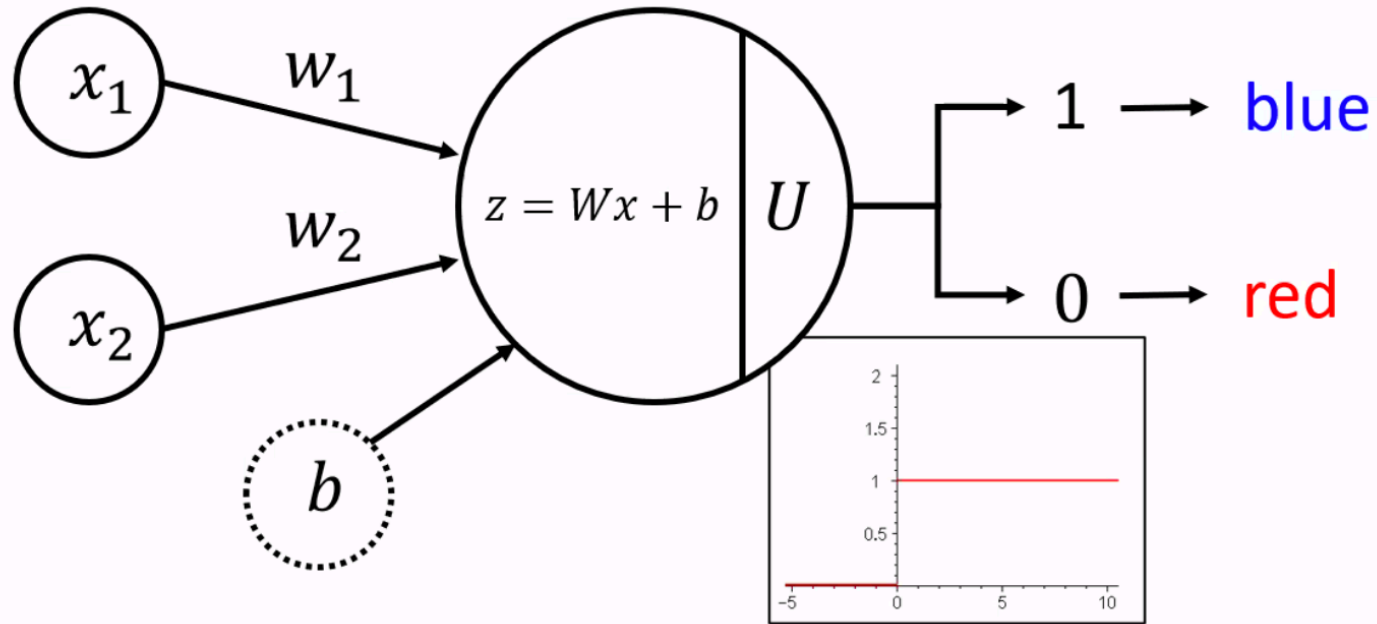
The AND operator

AND	x	y
No	0	0
No	1	0
No	0	1
Yes	1	1



The perceptron







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Vectors and Linear Transformations

Conclusion